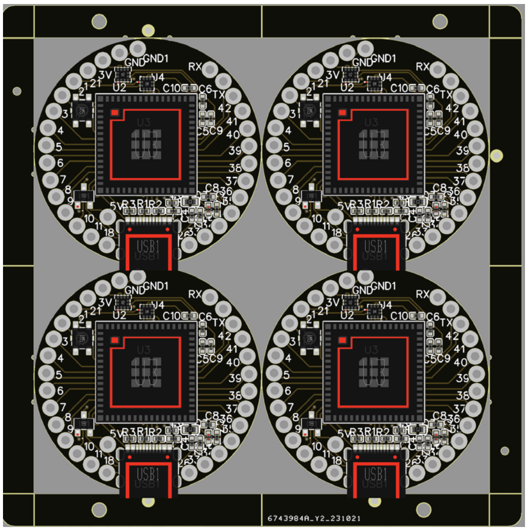
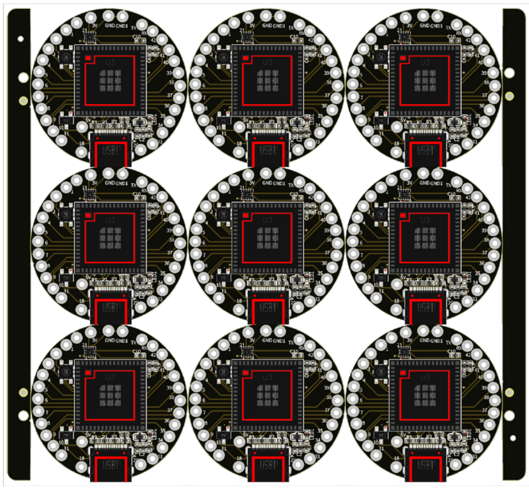


Edges and rails on all sides



Edges and rails on only two sides

Fig. 4: Panel production of PCBs

Panelising different shapes of PCBs. In some cases, different PCB shapes may be needed to design the product and its casing around the PCB. Panelising is also a useful method in such instances. However, when panelising different shapes, careful attention is required to the cutting slots and their placement within the panel.

Balancing copper distribution. Improper copper balance might make one side heavier, causing cracks and breaks, especially if the PCB is thin. Ensure even copper distribution to minimise warpage and thermal stress. Uneven copper can lead to bending and twisting during the soldering process.

Component placement. Avoid placing components too close to the edges of the PCBs. This precaution prevents damage during separation and ensures the integrity of the solder joints.

Tab routing for irregular shapes. For PCBs with highly irregular shapes, tab routing is an effective solution. Tabs are left between the PCBs, which can be broken or routed out after manufacturing. Ensure that tabs are placed in non-critical areas to avoid damaging components or traces.

Supporting structures. Use support structures like rails or frames to maintain panel stability during processing. These structures can be removed after the PCBs are separated.

Routing and V-scoring. Use routing and V-scoring techniques to separate the differently shaped PCBs. Routing cuts out the shapes, while V-scoring creates grooves for snapping the PCBs apart. Choose the method that best suits the complexity and size of the designs. For irregular shapes, stamp holes might be preferable, as V-cuts work best for rectangular PCBs.

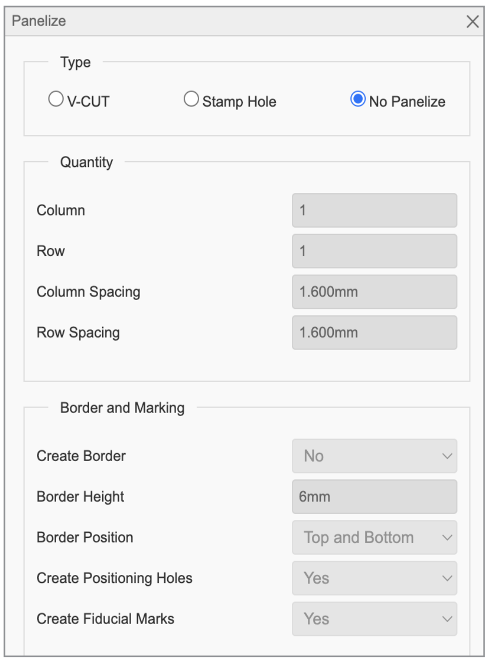


Fig. 5: Options available for snapping the PCB units from panel

Designing the panel for production

Once prototyping is complete, and the PCB designs are ready, costs can sometimes be reduced further by tweaking the panel layout to fit a few more PCB units. Here's how:

Tweaking edges and rails to save cost and adjust more PCBs in pan-